

9 Independence

Example 9.1. Do Americans think it is important for federal judges to be impartial in how they decide court cases? Consider the following data from the [Pew Research Center](#). (Independents are classified as either lean Democrat or lean Republican.)

	Democrat/lean Dem (D)	Republican/lean Rep (D^c)	Total
Important (I)	1674	1531	3205
Not important (I^c)	146	133	279
Total	1820	1664	3484

Suppose a person is randomly selected from this sample. Consider the events

$$I = \{\text{person thinks it is important for judges to be impartial}\}$$

$$D = \{\text{person is a Democrat/leans Dem}\}$$

1. Compute and interpret $P(I)$.
2. Compute and interpret $P(I|D)$.
3. Compute and interpret $P(I|D^c)$.

4. What do you notice about $P(I)$, $P(I|D)$, and $P(I|D^c)$?
5. Compute and interpret $P(D)$.
6. Compute and interpret $P(D|I)$.
7. Compute and interpret $P(D|I^c)$.
8. What do you notice about $P(D)$, $P(D|I)$, and $P(D|I^c)$?
9. Compute and interpret $P(D \cap I)$.
10. What is the relationship between $P(D \cap I)$ and $P(D)$ and $P(I)$?
11. When randomly selecting a person from this particular group, would you say that events

D and I are independent? Why?

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < P(A) < 1$ and $0 < P(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

A and B are independent

$$P(A \cap B) = P(A)P(B)$$

$$P(A^c \cap B) = P(A^c)P(B)$$

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$P(A|B) = P(A)$$

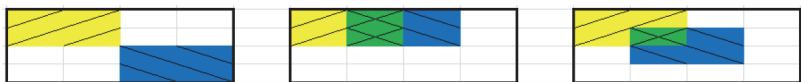
$$P(A|B) = P(A|B^c)$$

$$P(B|A) = P(B)$$

$$P(B|A) = P(B|A^c)$$

Example 9.2. Each of the three Venn diagrams below represents a sample space with 16 equally likely outcomes. Let A be the yellow / event, B the blue \ event, and their intersection $A \cap B$ the green $\times$ event. Suppose that areas represent probabilities, so that for example $P(A) = 4/16$.

In which of the scenarios are events A and B independent?



- Do not confuse “disjoint” with “independent”.

- Disjoint means two events do not “overlap”. Independence means two events “*overlap in just the right way*”.
- You can pretty much forget “disjoint” exists; you will naturally apply the addition rule for disjoint events correctly without even thinking about it.
- Independence is much more important and useful, but also requires more care.

Example 9.3. Roll two fair six-sided dice, one green and one gold. There are 36 total possible outcomes (roll on green, roll on gold), all equally likely. Consider the event $E = \{\text{the green die lands on 1}\}$. Answer the following questions by computing and comparing appropriate probabilities.

1. Consider $A = \{\text{the gold die lands on 6}\}$. Are A and E independent?

2. Consider $B = \{\text{the sum of the dice is 2}\}$. Are B and E independent?

3. Consider $C = \{\text{the sum of the dice is 7}\}$. Are C and E independent?

- Independence concerns whether or not the occurrence of one event affects the *probability* of the other.
- Given two events it is not always obvious whether or not they are independent.
- Independence depends on the underlying probability measure. Events that are independent under one probability measure might not be independent under another.
- Independence is often assumed. Whether or not independence is a valid assumption depends on the underlying random phenomenon.

Example 9.4. As in the birthday problem, assume that birthdays are distributed equally over 365 days. Randomly select 3 people.

- A_{12} is the event that the first and second persons selected have the same birthday

- A_{13} is the event that the first and third persons selected have the same birthday
- A_{23} is the event that the second and third persons selected have the same birthday

1. Are the two events A_{12} and A_{13} independent?

2. Are the two events A_{12} and A_{23} independent?

3. Are the two events A_{13} and A_{23} independent?

4. Are the three events A_{12} , A_{13} , and A_{23} independent?

- A collection of events $A_1, A_2, A_3, \dots$ are **independent** if:
 - any pair of events $A_i, A_j, (i \neq j)$ satisfies $P(A_i \cap A_j) = P(A_i)P(A_j)$,
 - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$,
 - and any quadruple of events satisfies $P(A_i \cap A_j \cap A_k \cap A_m) = P(A_i)P(A_j)P(A_k)P(A_m)$,
 - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.

Example 9.5. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.

2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.

3. Donny Don't says the answer to the previous part is $0.1 + 0.1 + 0.1 + 0.1 = 0.4$. Explain the error in Donny's reasoning.

- When events are independent, the multiplication rule simplifies greatly.

$$P(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n) = P(A_1)P(A_2)P(A_3) \cdots P(A_n) \quad \text{if } A_1, A_2, A_3, \dots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with “and” events whenever possible in order to use the multiplication rule.
- For example, for problems involving “at least one” (an “or” event) take the complement to obtain “none” (an “and” event).